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Public and Global Health: Role of Immunizations

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives:

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By the end of this lecture, you should be able to:

1. Explain how vaccines protect individuals and populations
2. Distinguish major vaccine types and platforms
3. Describe immune mechanisms of vaccine-induced protection
4. Compare efficacy vs effectiveness
5. Provide examples of vaccines preventing infectious and non-infectious diseases
6. Identify contraindications and common adverse events
7. Communicate with vaccine-hesitant patients effectively

- Connections
  - ScholarRx: Immunization

Source: Course Syllabus

List of Current U.S. Vaccines (partial)

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Influenza	IIV, LAIV, RIV	**Varicella**	VAR	**Mpox (monkeypox)**	MVA-BN (JYNNEOS)
Respiratory Syncytial Virus	RSV (RSVpreF)	**Zoster (Shingles)**	RZV (Shingrix)	**Smallpox/Monkeypox**	ACAM2000
Hepatitis A	HepA	**Pneumococcal conjugate**	PCV (PCV13, PCV15, PCV20)	**Rabies**	Rabies (PrEP/PEP)
Hepatitis B	HepB	**Pneumococcal polysaccharide**	PPSV23	**Yellow fever**	YF
Hepatitis A + B combination	HepA-HepB (Twinrix)	**Haemophilus influenzae type b**	Hib	**Typhoid**	Typh-I (injectable), Typh-O (oral)
Diphtheria, Tetanus, Pertussis	DTaP (peds)	**Meningococcal conjugate**	MenACWY	**Cholera**	CVD 103-HgR
Diphtheria, Tetanus, Pertussis	Tdap (adol/adult)	**Meningococcal serogroup B**	MenB	**Japanese encephalitis**	JE
Tetanus, Diphtheria	Td (booster)	**Meningococcal ACWY + B combo**	MenABCWY	**Tick-borne encephalitis**	TBE
Polio	IPV	**Human Papillomavirus**	HPV (9vHPV)	**Adenovirus (military)**	Ad4/Ad7
Measles, Mumps, Rubella	MMR	**Rotavirus**	RV (RV1, RV5)	**Anthrax**	AVA (special populations)
Measles, Mumps, Rubella + Varicella	MMRV	**Tuberculosis**	BCG (not routine U.S.)		

Why Vaccines Matter

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- Among the most effective interventions in medical history – prevents severe disease and transmission!
- Prevent morbidity, mortality, and disability resulting in the promotion of wellness throughout the lifespan
- Dramatic reduction healthcare costs
- Protect vulnerable populations who are unable to produce an immune response through herd effects
  - e.g., immune suppressed, contraindication to vaccines, refugees/ overcrowding, low-income countries
- Prevention or attenuation of outbreaks, epidemics, pandemics

Typical Course of Viral Infections

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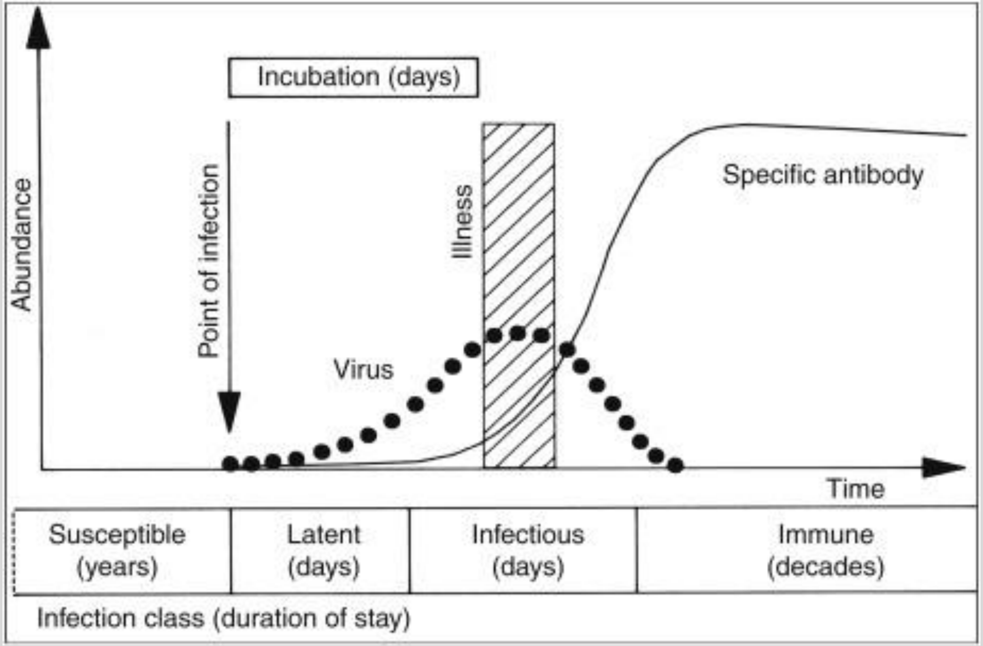

- Immunity occurs either via infection or immunization
- Primary infection without immunity results in disease and sequelae
  - Slow onset of primary response
  - Lower antibody titer
  - Initial IgM -> IgG class switch
- Transmission occurs until primary immunity controls infection

The Impact of Vaccination on the Epidemiology of Infectious Diseases. Roy M. Anderson The Vaccine Book, Chapter 1, 3-31

Vaccines Alter the Course of Infection

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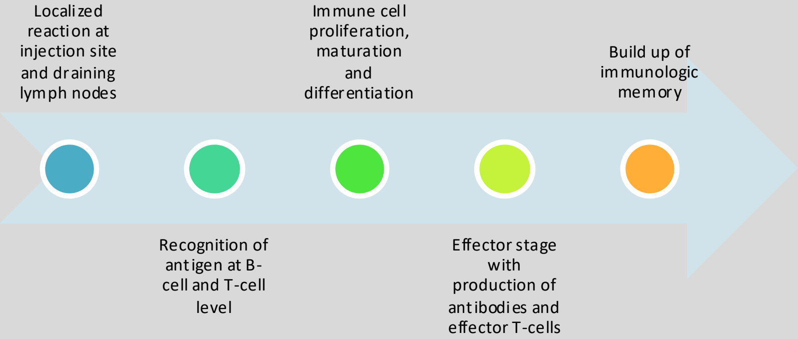

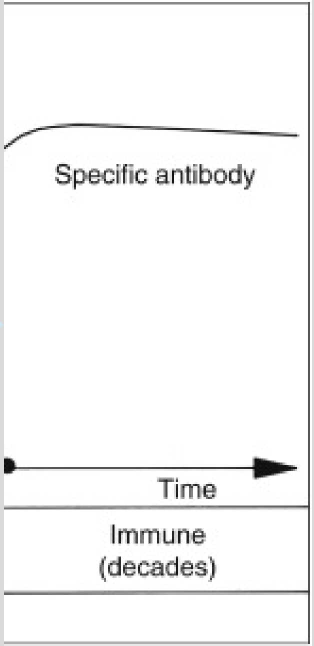

- Immunization primes the immune system to reduce or prevent disease, sequelae and transmission
- Secondary response
  - Rapid, more robust memory response
  - Higher affinity antibodies
  - Higher IgG titers

The Impact of Vaccination on the Epidemiology of Infectious Diseases. Roy M. Anderson The Vaccine Book, Chapter 1, 3-31

Herd Immunity

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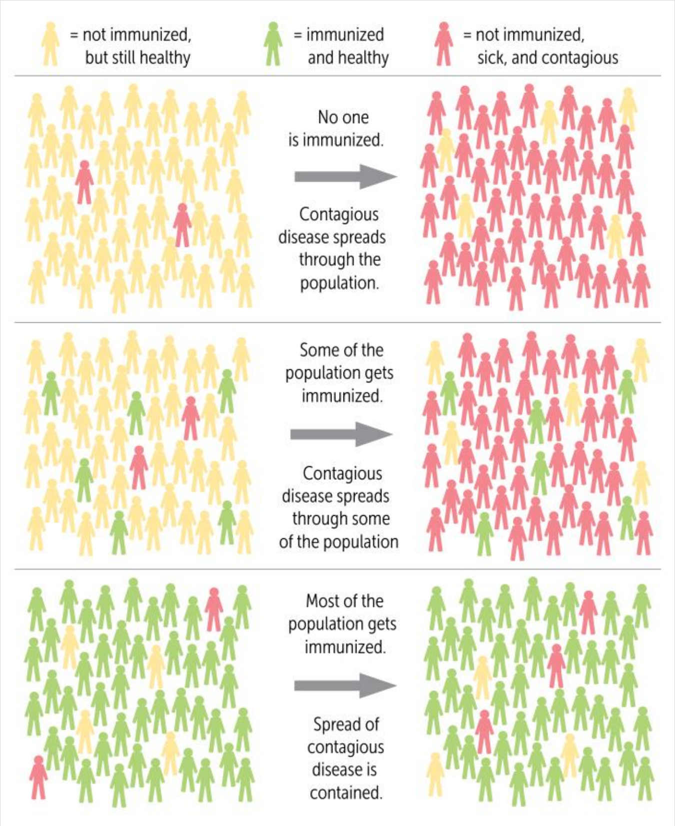
- The more people immune, the more likely transmission from an infectious person will hit a “dead end”
- Estimates of percentage of population immune to prevent outbreaks vary for disease (more contagious requires greater herd immunity)



Global Public Health Role of Vaccines

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- Reduced disease burden
  - ↓ Potential for pandemics and epidemics
  - ↓ Antimicrobial use/resistance burden
  - ↓ Infant and adult mortality and morbidity
  - ↓ Healthcare cost and burden
- Disease eradication
  - Smallpox, (almost) polio

Vaccine-preventable Complications

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- Diphtheria: upper airway obstruction, systemic toxin-mediated illness, myocarditis, arrhythmia, heart failure, peripheral neuropathy, renal injury HiB: otitis media, epiglottitis, meningitis, bacteremia, sepsis, septic arthritis, pneumonia, hearing loss, cognitive impairment/development delay, seizures
- HepA: fulminant hepatic failure (rare), cholestatic hepatitis
- HepB: chronic hepatitis, liver failure, cirrhosis, hepatocellular carcinoma, HepB-associated glomerulonephritis and polyarteritis nodosa
- HPV: recurrent respiratory papillomatosis, genital warts, cancers of cervix, anus, penis, oropharynx, vulva/vagina
- Influenza: viral pneumonia, secondary bacterial pneumonia, otitis media (children), myocarditis, encephalopathy, asthma exacerbation, COPD exacerbation, myocardial infarction, stroke
- Measles: viral pneumonia, secondary bacterial pneumonia, acute measles encephalitis, SSPE, post-measles immune suppression associated secondary infection
- Meningococcus: meningitis, septic shock, purpura fulminans and limb loss, hearing loss, permanent neurologic impairment, chronic kidney injury
- Mumps: aseptic meningitis, orchitis, infertility, oophoritis, mastitis, sensorineural hearing loss
- Pertussis: lower respiratory infection, apnea and hypoxic brain injury (infants), pneumonia, seizures, encephalopathy
- Pneumococcal: invasive pneumonia, sepsis, septic shock, meningitis, otitis media (children), hearing loss
- Polio: acute poliomyelitis, irreversible flaccid paralysis, respiratory failure due to respiratory muscle paralysis, post-polio syndrome
- Rotavirus: dehydration, shock, electrolyte derangements
- RSV: hypoxic respiratory failure, asthma exacerbation, reductions in all-cause mortality in older adults
- Rubella: congenital rubella syndrome (cataracts, deafness, patent ductus arteriosus, pulmonary artery stenosis, microcephaly, developmental delay)
- SARS-CoV-2: carditis, long-COVID
- Tetanus: toxin-mediated spastic paralysis, layngospasm and respiratory failure, autonomic instability, aspiration pneumonia, fractures, rhabdomyolysis
- Varicella: viral pneumonia, bacterial skin superinfection, necrotizing fasciitis, encephalitis, cerebellitis, herpes zoster, postherpetic neuralgia, vision loss (herpes zoster ophthalmicus)

Vaccine-preventable Complications

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- **MMR** → prevents encephalitis + SSPE + congenital rubella syndrome
- **DTaP/Tdap** → prevents airway obstruction (diphtheria), spasms (tetanus), hypoxic brain injury (pertussis)
- **Hib/PCV/MenACWY/MenB** → prevents meningitis → prevents hearing loss + neurologic disability
- **IPV** → prevents irreversible paralysis, post-polio syndrome
- **HepB** → prevents chronic hepatitis → cirrhosis → HCC
- **HPV** → prevents persistent infection → cervical/oropharyngeal cancer
- **Rotavirus** → prevents severe dehydration hospitalization/death
- **Influenza/COVID** → prevents severe pneumonia, death, and reduces cardiovascular/inflammatory sequelae
- **Shingrix** → prevents shingles → postherpetic neuralgia

Platforms Conferring Immunity

Vaccines (active)

- Live attenuated
- Inactivated (killed)
- Subunit/recombinant protein
- Toxoid
- Polysaccharide
- Conjugate
- mRNA
- Viral vector

Passive immunity

- Instant effect, poor durability, no effect on adaptive immunity
- Immunoglobulin preparations
  - Maternal IgG transfer
  - IVIG
  - Rabies immune globulin
  - HBIG
- Monoclonal antibodies
  - Palivizumab (RSV)
  - Casirivimab/imdevimab (SARS-CoV-2)

Live Attenuated & Inactivated Vaccines

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Live attenuated

Examples: MMR, varicella, rotavirus, LAIV, YF, OPV, oral typhoid

- Replication of intact, weakened pathogen in host
- Can be cold-adapted (LAIV)
- Strong, long-lasting humoral and cellular immunity
- Contraindicated in pregnancy and immune compromised
  - (e.g., low CD4, transplant, cytotoxic chemotherapy)

Inactivated

Examples: IPV, hepA, rabies, inactivated influenza, whole killed pertussis

- Heat, radiation, chemical inactivation
- No intact pathogen = no replication
- Strong humoral and T-cell immunity
- Long-lasting immunity requires boosters/adjuvants
- Can be given in pregnancy and to immune compromised

Subunit & Toxoid Vaccines

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Subunit

Examples: HepB, Shingrix, Novavax COVID

- Only specific antigens included
- Proteins or glycoproteins
- Manufactured from genetic code in producer cell
- Require adjuvants/boosters
- Excellent safety profile

Toxoid

Examples: tetanus, diphtheria

- Inactivated bacterial toxin
- Prevents toxin-mediated disease
- Primarily humoral immunity to toxin
- Bacterial growth unaffected
- Requires boosters

Polysaccharide & Conjugate Vaccines

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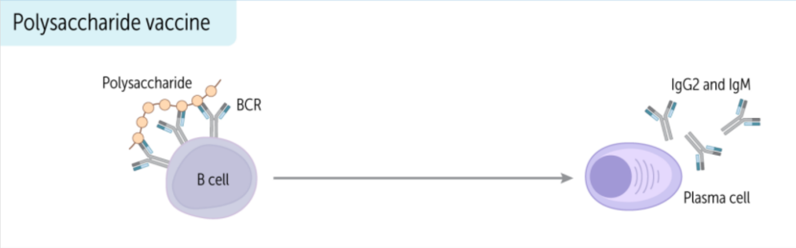
Polysaccharide

Examples: PPSV23

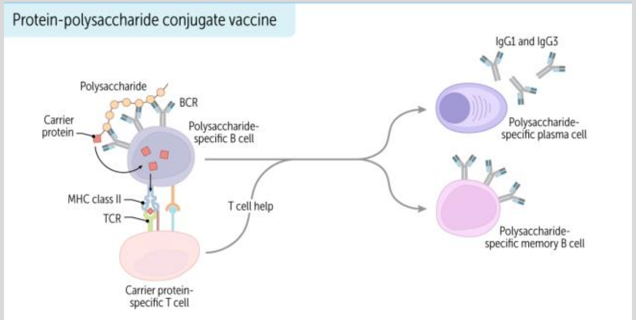
- Capsule polysaccharides
- Provides immunity to many serotypes
- Poor memory in infants
- Poor T-cell activation
- Less durable immunity



Conjugate

Examples: Hib, PCV13/15/20, MenACWY

- Polysaccharide + protein carrier
- Converts response to T-dependent
- Effective in infants
- Durable memory response



mRNA & Viral Vector Vaccines

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mRNA

Examples: Moderna/Pfizer COVID-19, Moderna CMV

- Lipid nanoparticle carrier delivers mRNA segment to local lymph nodes
- Antigen made rapidly in host cell cytoplasm
- Strong, durable antibody + T-cell response
- Low manufacturing cost
- Rare carditis

Viral Vector

Examples: J&J COVID, Ebola

- Modified, harmless virus delivers antigen gene
- Strong cellular immunity
- Pre-existing immunity to vector may reduce effectiveness
- Rare thrombosis with thrombocytopenia syndrome

Vaccine Contraindications

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- Severe allergic reaction (eg, anaphylaxis) to a component of the vaccine
- Known severe immunodeficiency for patients and their close contacts (live attenuated vaccines)
- History of intussusception (rotavirus vaccine)
- Pregnancy (varicella vaccine, live-attenuated influenza vaccine, MMR)

Side Effects of Common Vaccines

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Vaccine	Common Side Effects (Mild)	Rare or Serious Adverse Events
Influenza (Flu)	Soreness/redness at injection site, low-grade fever, muscle aches, headache.	Guillain-Barré Syndrome (GBS) (approx. 1-2 cases per million); Anaphylaxis.
MMR (Measles, Mumps, Rubella)	Fever, mild rash, swollen glands in cheeks or neck.	Febrile seizures (approx. 1 in 3,000); Temporary low platelet count.
DTaP / Tdap (Tetanus, Diptheria, Pertussis)	Pain/swelling at site, fussiness, tiredness, loss of appetite.	Permanent brain damage (extremely rare, link is debated); Severe swelling of the entire limb.
HPV (Gardasil 9)	Pain/redness at site, headache, nausea, dizziness.	Syncope (fainting)—especially in adolescents; clinical protocol suggests 15-min observation.
Varicella (Chickenpox)	Sore arm, mild rash (up to 1 month after), joint stiffness.	Pneumonia; Seizures (very rare).
Hepatitis B	Soreness at injection site, temperature of 99.9° F or higher.	Severe allergic reaction (anaphylaxis) to yeast (used in production).
Shingles (Shingrix)	Shivering, muscle pain, fatigue, upset stomach.	Guillain-Barré Syndrome (GBS) (rare risk in seniors).
Pneumococcal (PCV15/20)	Swelling/redness at site, loss of appetite, irritability.	High fever; Allergic reactions to vaccine components.

Efficacy versus Effectiveness

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- **Efficacy:** reduction of disease in controlled trials
- **Effectiveness:** real-world performance (population-based)

- **Why They Differ**
  - adherence to dosing schedule
  - storage/handling errors
  - host immune status
  - pathogen evolution
  - socioeconomic barriers

- **Outcomes Measured**
  - infection prevention
  - reduction in severe disease
  - prevention of complications
  - reduction in transmission

Influenza Vaccine

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- Inactivated or recombinant (typically)
- Neutralizing antibodies against hemagglutinin (HA) prevent cell entry preventing reduce infection severity
- Protection varies due to antigenic drift; is absent after antigenic shift
- Expect 50-70% effectiveness

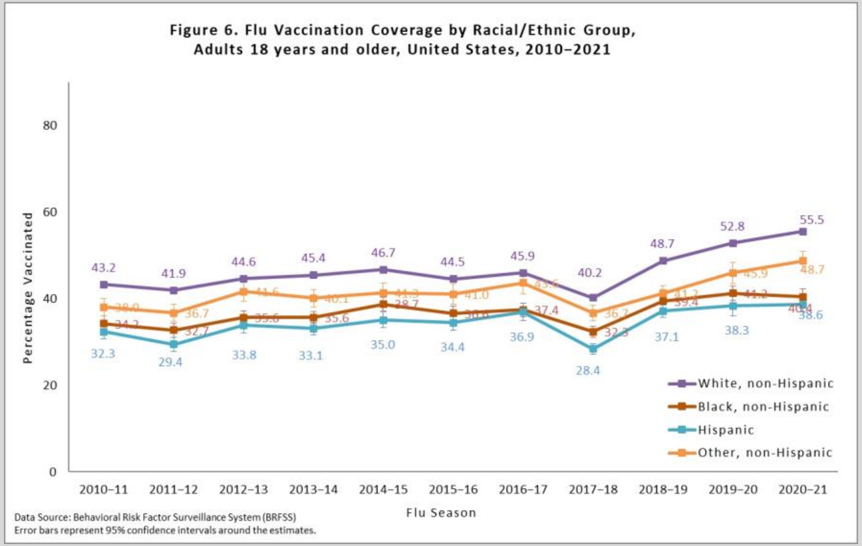

Figure 6. Flu Vaccination Coverage by Racial/Ethnic Group, Adults 18 years and older, United States, 2010–2021

Data Source: Behavioral Risk Factor Surveillance System (BRFSS)
Error bars represent 95% confidence intervals around the estimates.

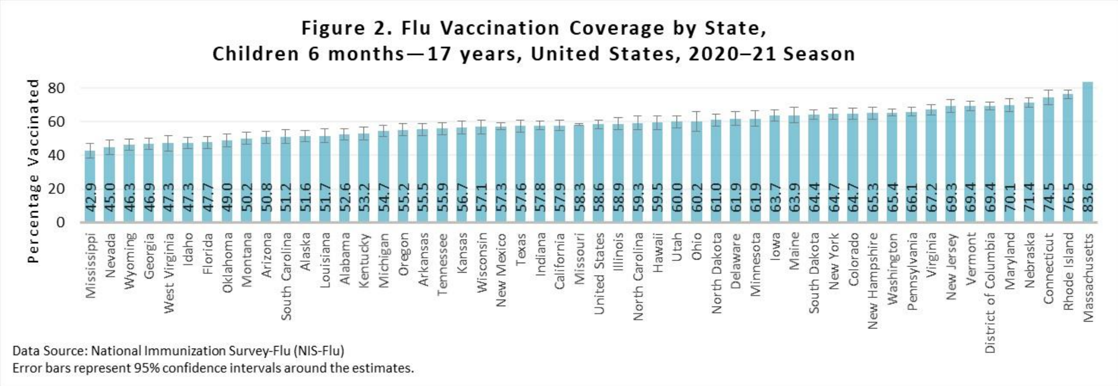

Figure 2. Flu Vaccination Coverage by State, Children 6 months—17 years, United States, 2020–21 Season

Data Source: National Immunization Survey-Flu (NIS-Flu)
Error bars represent 95% confidence intervals around the estimates.

CDC

Measles Vaccine

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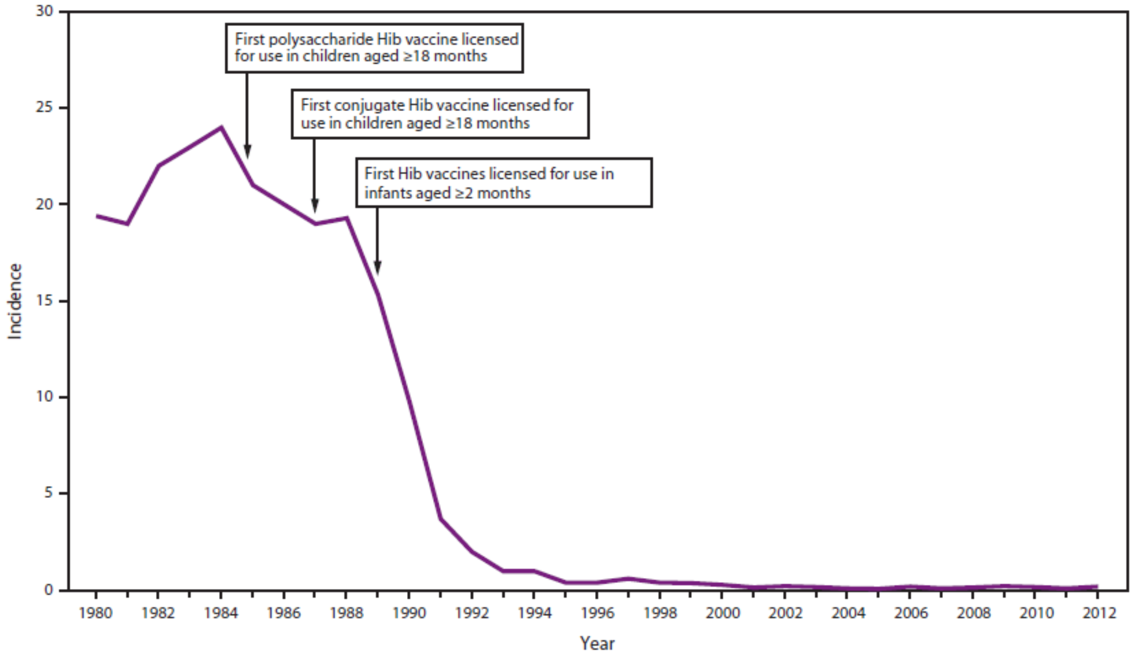
- Live attenuated virus that induced neutralizing antibodies + memory T cells
- Highly effective after 2 doses
- High herd immunity threshold (95%)
- Prevented 23 million deaths between 2010 and 2018 (WHO)
  - Measles pneumonia, SSPE, post-measles immune suppression

Haemophilus influenzae type B (Hib)



- Within years of approval, 99% reduction in invasive disease (20,000 to few)
- Reduction in transmission in unvaccinated (herd effect)
- Remaining cases mostly in adults, non-type B in children

Pneumococcal Vaccine

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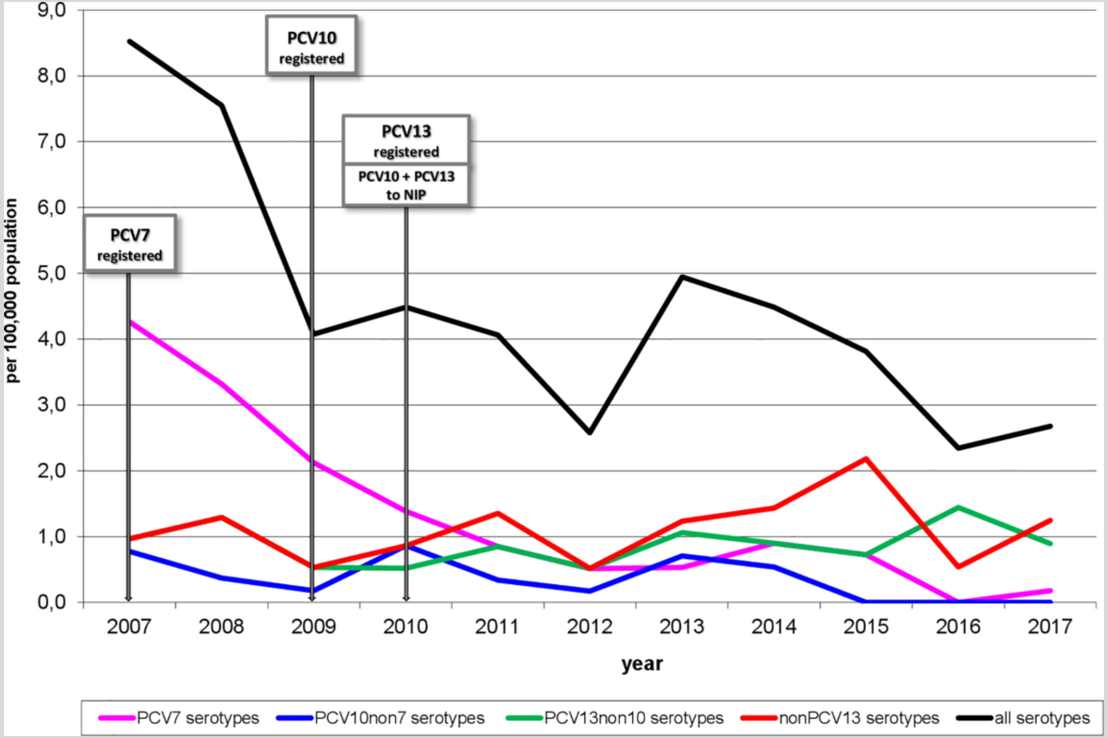

- 45% ↓ in hospitalizations
- Children 68% ↓ in invasive disease, ~75% ↓ mortality

Hepatitis B Vaccine

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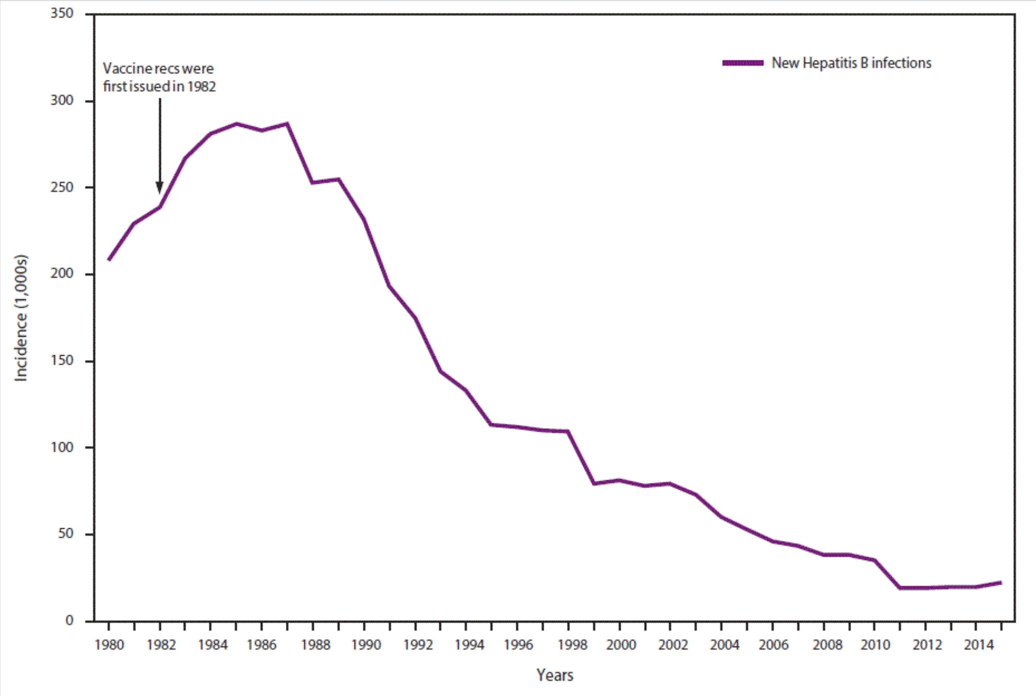

- Near-total elimination of mother-to-child transmission with newborn dose
- 99% ↓ peds prevalence
- Estimated 39 million lives saved for years 2000-2030
- Probably life-long immunity
- 62-70% ↓ HCC mortality

Human Papillomavirus Vaccine

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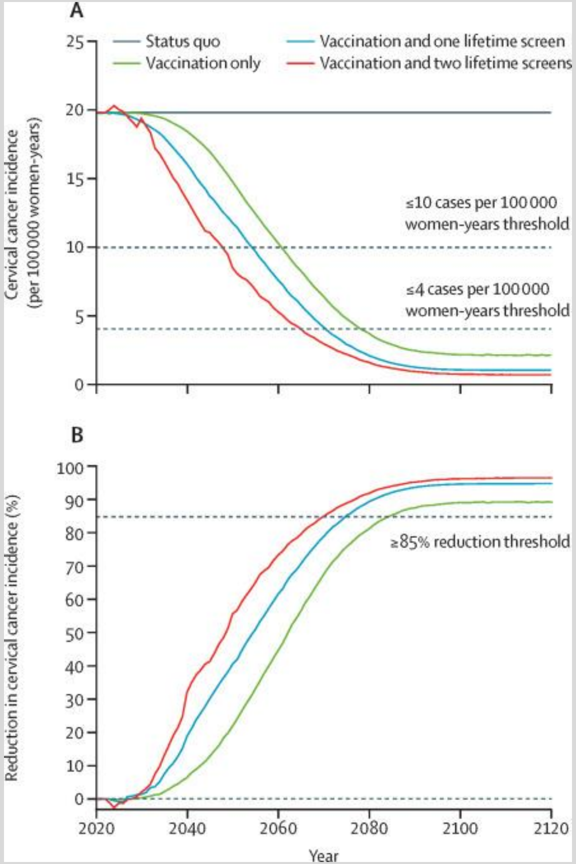

- 56-71% ↓ HPV prevalence among adolescent and young women
- 88% ↓ warts teen girls
- Significant herd effect
- 80-90% ↓ cervical cancer when given before 17y

COVID-19 (SARS-CoV-2)

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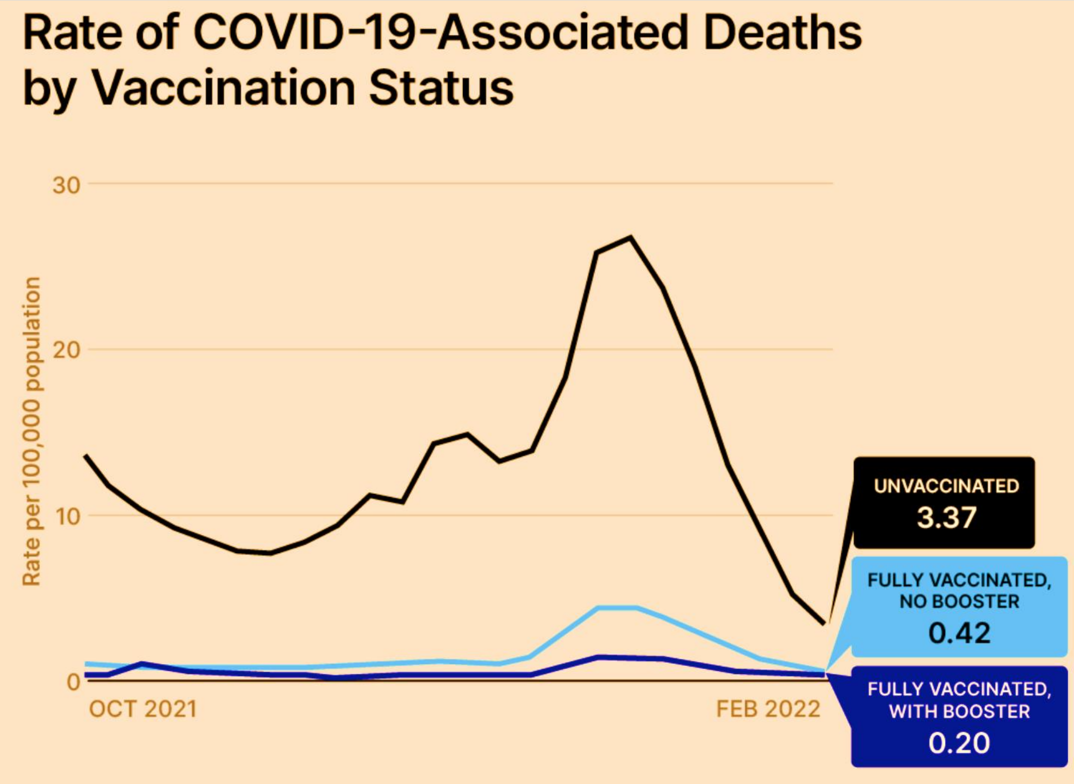

Source: CDC - https://www.cdc.gov/coronavirus/2019-ncov/images/communication/covid-data-tracker/321563-EO_COVID-Data-Tracker_Vaccine-and-Booster-Demographics_1080x1080.png

Determining Who and When to Vaccinate

- U.S.: AAAP, IDSA, CDC
- Globally, foreign travel: WHO, CDC

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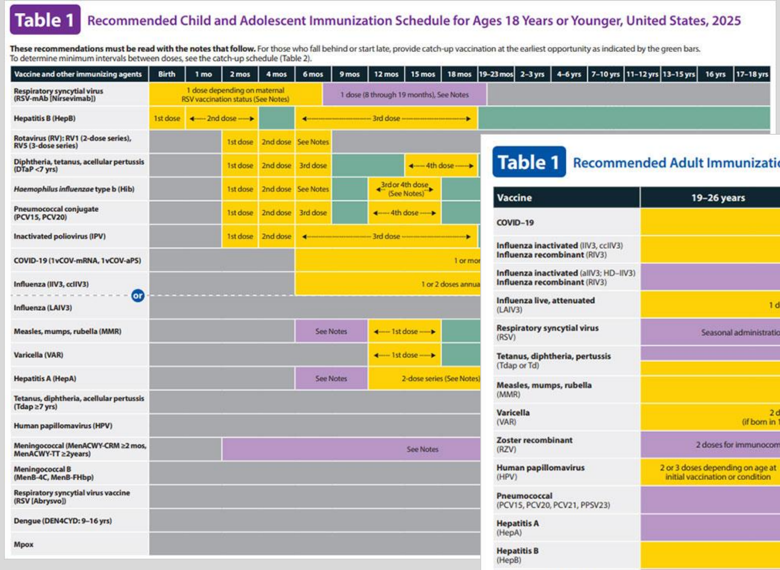


Table 1 Recommended Child and Adolescent Immunization Schedule for Ages 18 Years or Younger, United States, 2025

These recommendations must be read with the notes that follow. For those who fall behind or start late, provide catch-up vaccination at the earliest opportunity as indicated by the green bars.
To determine minimum intervals between doses, see the catch-up schedule (Table 2).

Vaccine and other immunizing agents	Birth	1 mo	2 mos	4 mos	6 mos	9 mos	12 mos	15 mos	18 mos	19–23 mos	2–3 yrs	4–6 yrs	7–10 yrs	11–12 yrs	13–15 yrs	16 yrs	17–18 yrs
Respiratory syncytial virus (RSV-mAb [Nirsevimab])	1 dose depending on maternal RSV vaccination status (See Notes)					1 dose (8 through 19 months), See Notes											
Hepatitis B (HepB)	1st dose	←2nd dose→			←3rd dose→												
Rotavirus (RV): RV1 (2-dose series), RV5 (3-dose series)			1st dose	2nd dose	See Notes												
Diphtheria, tetanus, acellular pertussis (DTaP <7 yrs)			1st dose	2nd dose	3rd dose			←4th dose→									
Haemophilus influenzae type b (Hib)			1st dose	2nd dose	See Notes		←3rd or 4th dose (See Notes)→										
Pneumococcal conjugate (PCV15, PCV20)			1st dose	2nd dose	3rd dose		←4th dose→										
Inactivated poliovirus (IPV)			1st dose	2nd dose	←3rd dose→												
COVID-19 (1vCOV-mRNA, 1vCOV-aPS)					1 or mor												
Influenza (IIV3, ccIIV3) or					1 or 2 doses annua												
Influenza (LAIV3)																	
Measles, mumps, rubella (MMR)					See Notes		←1st dose→										
Varicella (VAR)							←1st dose→										
Hepatitis A (HepA)					See Notes		2-dose series (See Notes)										
Tetanus, diphtheria, acellular pertussis (Tdap ≥7 yrs)																	
Human papillomavirus (HPV)																	
Meningococcal (MenACWY-CRM ≥2 mos, MenACWY-TT ≥2years)			See Notes														
Meningococcal B (MenB-4C, MenB-FHbp)																	
Respiratory syncytial virus vaccine (RSV [Abrysvo])																	
Dengue (DEN4CYD: 9–16 yrs)																	
Mpox																	

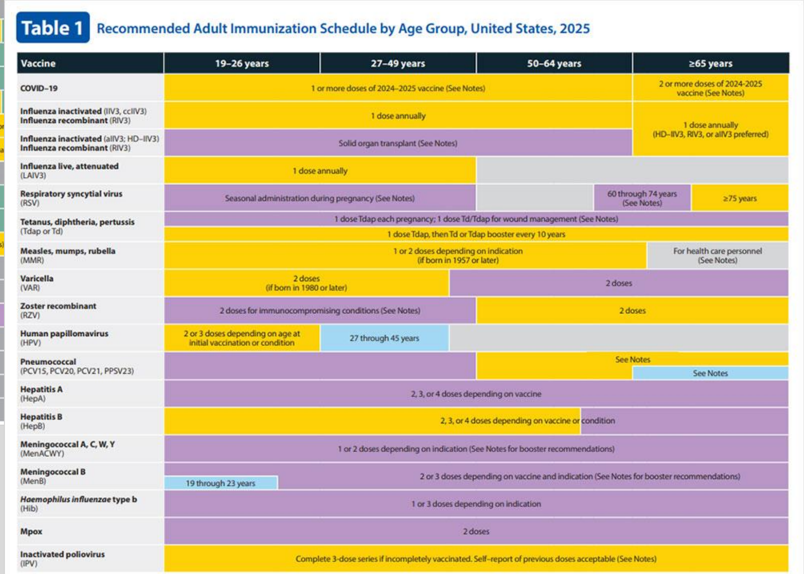
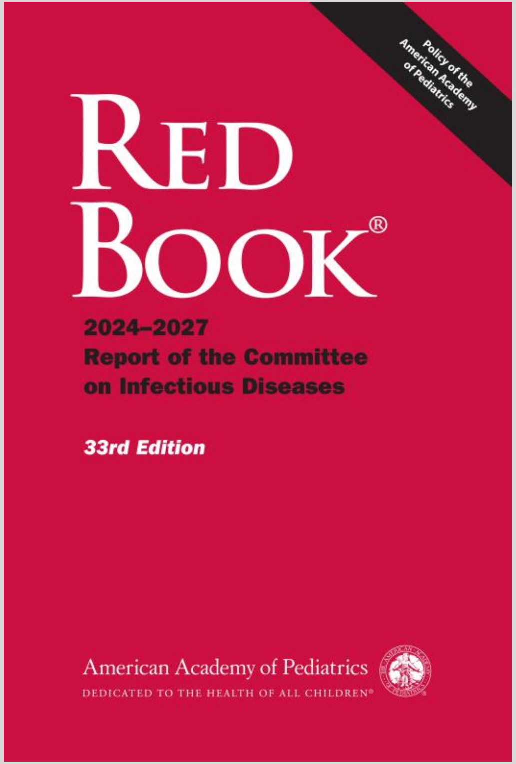


Table 1 Recommended Adult Immunization Schedule by Age Group, United States, 2025

Vaccine	19–26 years	27–49 years	50–64 years	≥65 years
COVID–19	1 or more doses of 2024–2025 vaccine (See Notes)			2 or more doses of 2024-2025 vaccine (See Notes)
Influenza inactivated (IIV3, ccIIV3) **Influenza recombinant** (RIV3)	1 dose annually			1 dose annually (HD–IIV3, RIV3, or aIIV3 preferred)
Influenza inactivated (aIIV3; HD–IIV3) **Influenza recombinant** (RIV3)	Solid organ transplant (See Notes)			
Influenza live, attenuated (LAIV3)	1 dose annually			
Respiratory syncytial virus (RSV)	Seasonal administration during pregnancy (See Notes)		60 through 74 years (See Notes)	≥75 years
Tetanus, diphtheria, pertussis (Tdap or Td)	1 dose Tdap each pregnancy; 1 dose Td/Tdap for wound management (See Notes) 1 dose Tdap, then Td or Tdap booster every 10 years			
Measles, mumps, rubella (MMR)	1 or 2 doses depending on indication (if born in 1957 or later)			For health care personnel (See Notes)
Varicella (VAR)	2 doses (if born in 1980 or later)	2 doses		
Zoster recombinant (RZV)	2 doses for immunocompromising conditions (See Notes)		2 doses	
Human papillomavirus (HPV)	2 or 3 doses depending on age at initial vaccination or condition	27 through 45 years		
Pneumococcal (PCV15, PCV20, PCV21, PPSV23)			See Notes	See Notes
Hepatitis A (HepA)	2, 3, or 4 doses depending on vaccine			
Hepatitis B (HepB)	2, 3, or 4 doses depending on vaccine or condition			
Meningococcal A, C, W, Y (MenACWY)	1 or 2 doses depending on indication (See Notes for booster recommendations)			
Meningococcal B (MenB)	19 through 23 years 2 or 3 doses depending on vaccine and indication (See Notes for booster recommendations)			
***Haemophilus influenzae* type b** (Hib)	1 or 3 doses depending on indication			
Mpox	2 doses			
Inactivated poliovirus (IPV)	Complete 3-dose series if incompletely vaccinated. Self-report of previous doses acceptable (See Notes)			



Public Perception of Vaccines

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- Vaccine
  - Acceptance – take recommended vaccines
  - Hesitant – want to to what is best for their family and themselves
  - Anti-vaxxer – alternative motives, secondary gain
- Who can I trust?
- Alternative vaccine schedules?

Causes of non-Vaccine Acceptance

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- Lack of personal relevance
- Lack of understanding (statistics, study design)
- Lack of experience
- Misinterpretation (Vaccine Dragon*)
- Lack of trust
- Cultural
- Rumor/family and social networks
- Self-validating isolationism
- Social media algorithms
- Opportunities for messaging
- Misinformation/secondary gain


A Newsreel of Disaster!

FULL PAGE OF WIREPHOTOS ON PAGE B
A Moment-to-Moment Story of the Blast

San Francisco Chronicle
The City's Only Home-Owned Newspaper

Hindenburg, World's Largest Zeppelin, Destroyed in Lakehurst Explosion; 33 Killed, 64 Miraculously Escape

Photo Made at Instant of Blast

$1000 IN YOUR PAY ENVELOPE

Crowd Views Landing Field Catastrophe

HITLER STUNNED BY TRAGEDY

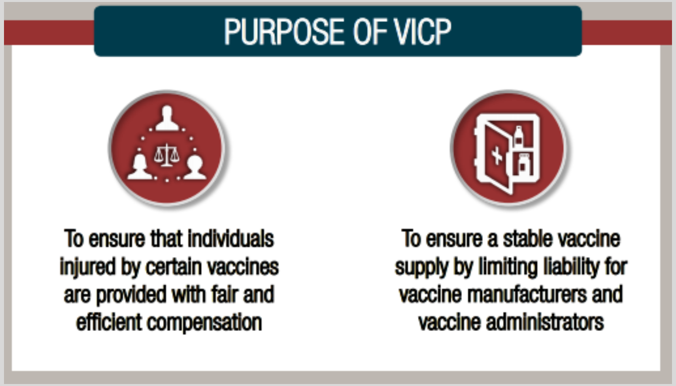
*e.g., "I get the flu every time I get the flu vaccine"

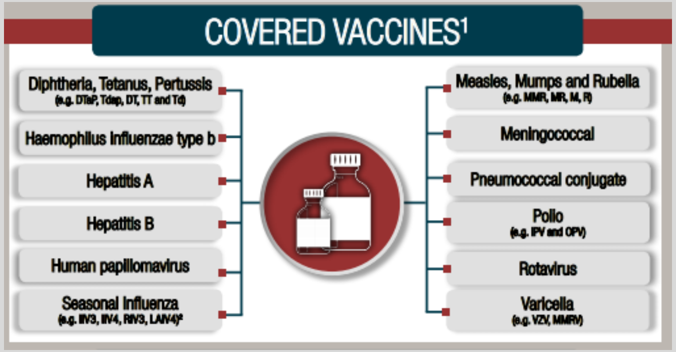
Vaccine Truths

1. Vaccines do not prevent infection, they reduce the risk of severe disease, death, sequelae and complications
2. Inactivated or component vaccines do not replicate or cause disease
3. Some vaccines result in immune protection that is more robust than infection
4. Current vaccines were either studied in placebo-controlled trials or were studied in comparison to vaccines that were
5. With exception of SARS-CoV-2 (the pandemic drove economic development), vaccines for infections do not provide significant revenue for pharma
6. Side effects of vaccines, in general, are less prevalent (sometimes dramatically) than similar effects of infection
7. The effectiveness of unstudied, alternate schedules cannot be predicted by studied schedules
8. Eradication of disease does not eradicate circulating virus in all cases (e.g., polio)
9. Antigen exposure from vaccine is extremely limited (1-5) compared to the thousands of exposures of daily life
10. Vaccines do not enter the host cell nucleus or alter DNA; infectious agents do
11. No currently U.S. licensed vaccine contains aluminum, nor has aluminum in vaccines been linked to any disease process
12. No currently U.S. licensed vaccine contains thimerosal, not has thimerosal in vaccines been linked to any disease process
13. Vaccines do not promote promiscuity or dangerous behavior
14. Anticancer vaccines are now being introduced for treatment

National Vaccine Injury Compensation Program

- No-fault government funded program
- Created in 1980's as mode to compensate people injured by vaccines





HRSA

National Vaccine Injury Compensation Program

- Since 1988, ~$4.1 billion in compensation
- 1 compensation for every 1 million doses of vaccines

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Fiscal Year	Number of Compensated Awards	Petitioners' Award Amount	Attorneys' Fees/Costs Payments	Number of Payments to Attorneys (Dismissed Cases)	Attorneys' Fees/Costs Payments (Dismissed Cases)	Number of Payments to Interim Attorneys'	Interim Attorneys' Fees/Costs Payments	Total Outlays
FY 1989	6	$1,317,654.78	$54,107.14	0	$0.00	0	$0.00	$1,371,761.92
FY 1990	88	$53,252,510.46	$1,379,005.79	4	$57,699.48	0	$0.00	$54,689,215.73
FY 1991	114	$95,980,493.16	$2,364,758.91	30	$496,809.21	0	$0.00	$98,842,061.28
FY 1992	130	$94,538,071.30	$3,001,927.97	118	$1,212,677.14	0	$0.00	$98,752,676.41
FY 1993	162	$119,693,267.87	$3,262,453.06	272	$2,447,273.05	0	$0.00	$125,402,993.98
FY 1994	158	$98,151,900.08	$3,571,179.67	335	$3,166,527.38	0	$0.00	$104,889,607.13
FY 1995	169	$104,085,265.72	$3,652,770.57	221	$2,276,136.32	0	$0.00	$110,014,172.61
FY 1996	163	$100,425,325.22	$3,096,231.96	216	$2,364,122.71	0	$0.00	$105,885,679.89
FY 1997	179	$113,620,171.68	$3,898,284.77	142	$1,879,418.14	0	$0.00	$119,397,874.59
FY 1998	165	$127,546,009.19	$4,002,278.55	121	$1,936,065.50	0	$0.00	$133,484,353.24
FY 1999	96	$95,917,680.51	$2,799,910.85	117	$2,306,957.40	0	$0.00	$101,024,548.76
FY 2000	136	$125,945,195.64	$4,112,369.02	80	$1,724,451.08	0	$0.00	$131,782,015.74
FY 2001	97	$105,878,632.57	$3,373,865.88	57	$2,066,224.67	0	$0.00	$111,318,723.12
FY 2002	80	$59,799,604.39	$2,653,598.89	50	$656,244.79	0	$0.00	$63,109,448.07
FY 2003	65	$82,816,240.07	$3,147,755.12	69	$1,545,654.87	0	$0.00	$87,509,650.06
FY 2004	57	$61,933,764.20	$3,079,328.55	69	$1,198,615.96	0	$0.00	$66,211,708.71
FY 2005	64	$55,065,797.01	$2,694,664.03	71	$1,790,587.29	0	$0.00	$59,551,048.33
FY 2006	68	$48,746,162.74	$2,441,199.02	54	$1,353,632.61	0	$0.00	$52,540,994.37
FY 2007	82	$91,449,433.89	$4,034,154.37	61	$1,692,020.25	0	$0.00	$97,175,608.51
FY 2008	141	$75,716,552.06	$5,191,770.83	74	$2,531,394.20	2	$117,265.31	$83,556,982.40
FY 2009	131	$74,142,490.58	$5,404,711.98	36	$1,557,139.53	28	$4,241,362.55	$85,345,704.64
FY 2010	173	$179,387,341.30	$5,961,744.40	59	$1,933,550.09	22	$1,978,803.88	$189,261,439.67
FY 2011	251	$216,319,428.47	$9,572,042.87	403	$5,589,417.19	28	$2,001,770.91	$233,482,659.44
FY 2012	249	$163,491,998.82	$9,241,427.33	1,020	$8,649,676.56	37	$5,420,257.99	$186,803,360.70
FY 2013	375	$254,666,326.70	$13,543,099.70	704	$7,012,615.42	50	$1,454,851.74	$276,676,893.56
FY 2014	365	$202,084,196.12	$12,161,422.64	508	$6,824,566.68	38	$2,493,460.73	$223,563,646.17
FY 2015	508	$204,137,880.22	$14,445,776.29	118	$3,546,785.14	50	$3,089,497.68	$225,219,939.33
FY 2016	689	$230,140,251.20	$16,225,881.12	99	$2,741,830.10	59	$3,502,709.91	$252,610,672.33
FY 2017	706	$252,245,932.78	$22,045,785.00	131	$4,441,724.32	52	$3,363,464.24	$282,096,906.34
FY 2018	522	$199,658,492.49	$16,658,440.14	111	$5,091,269.45	58	$5,220,096.78	$226,628,298.86
FY 2019	276	$119,344,851.56	$8,091,796.70	41	$1,985,117.67	30	$2,064,009.07	$131,485,775.00
Total	**6,465**	**$3,807,498,922.78**	**$195,163,743.12**	**5,391**	**$82,076,204.20**	**454**	**$34,947,550.79**	**$4,119,686,42.89**

Pro:

No-fault, covers legal fees

60% cases settled

Funded by $0.75/vaccine

Con:

Long wait times

Case dismissal rate high

Lack of funding as vaccination rates fall

Only 60% of cases settled

National VICP Monthly Statistics

Ethical Considerations of Vaccination

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- Herd immunity to protect those unvaccinated, immune suppressed
- Continuity of care for vaccine-hesitant, anti-vax
- Opportunities to educate
- OPV in low-income settings

Thank you for reviewing and good luck!

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- Summary
  - Vaccines create memory and prevent severe outcomes
  - Different vaccine types create different immune responses
  - Vaccines protect individuals and communities
  - Vaccines prevent infectious AND non-infectious disease outcomes
  - Clinicians must understand immunology and communicate effective
- Feedback